# Database Design – TeamMatch

## 1. Entities

The main entities for TeamMatch are:

- Student
- Course
- Instructor
- Team
- TeamMember
- MatchRun
- StudentProfile

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## 2. Attributes

## Student

| Attribute | Key | Description |
|---|---|---|
| student_id | PK | Unique student ID |
| name | | Student name |
| email | | Student email |
| course_id | FK | Course the student belongs to |

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## Instructor

| Attribute | Key | Description |
|---|---|---|
| instructor_id | PK | Unique instructor ID |
| name | | Instructor name |
| email | | Instructor email |

## Course

| Attribute | Key | Description |
|---|---|---|
| course_id | PK | Unique course ID |
| course_name | | Name of the course |
| instructor_id | FK | Instructor assigned to course |
| team_size | | Required team size |

## StudentProfile

| Attribute | Key | Description |
|---|---|---|
| profile_id | PK | Unique profile ID |
| student_id | FK | Student linked to profile |
| skills | | Student skills, such as Python or frontend |
| availability | | Student available days/times |
| preferred_role | | Preferred role on a team |

## Team

| Attribute | Key | Description |
|---|---|---|
| team_id | PK | Unique team ID |
| course_id | FK | Course the team belongs to |
| run_id | FK | Match run that created the team |
| team_score | | Score calculated by matching algorithm |

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## TeamMember

| Attribute | Key | Description |
|---|---|---|
| team_id | PK, FK | Team ID |
| student_id | PK, FK | Student ID |

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## MatchRun

| Attribute | Key | Description |
|---|---|---|
| run_id | PK | Unique match run ID |
| course_id | FK | Course being matched |
| created_at | | Time the match was generated |
| status | | Status such as completed or failed |

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# 3. Primary Keys

- Student.student_id
- Instructor.instructor_id
- Course.course_id
- StudentProfile.profile_id
- Team.team_id
- MatchRun.run_id
- TeamMember.team_id + TeamMember.student_id

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# 4. Foreign Keys

- Student.course_id → Course.course_id
- Course.instructor_id → Instructor.instructor_id

- StudentProfile.student_id → Student.student_id
- Team.course_id → Course.course_id
- Team.run_id → MatchRun.run_id
- TeamMember.team_id → Team.team_id
- TeamMember.student_id → Student.student_id
- MatchRun.course_id → Course.course_id

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# 5. Relationships

- One Instructor can teach many Courses. **1:N**
- One Course can have many Students. **1:N**
- One Student has one StudentProfile. **1:1**
- One Course can have many MatchRuns. **1:N**
- One MatchRun can create many Teams. **1:N**
- One Course can have many Teams. **1:N**
- One Team can have many Students, and one Student can belong to a Team through TeamMember. **M:N resolved using TeamMember**

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# 6. ER Diagram

Copy this into your document as the ER diagram text, or recreate it in diagrams.net.

Instructor (1) ————————< (N) Course

Course (1) ————————————< (N) Student

Student (1) ———————————— (1) StudentProfile

Course (1) ————————————< (N) MatchRun

MatchRun (1) ———————————< (N) Team

Course (1) ————————————< (N) Team

Student (N) ————————< TeamMember >———————— (N) Team

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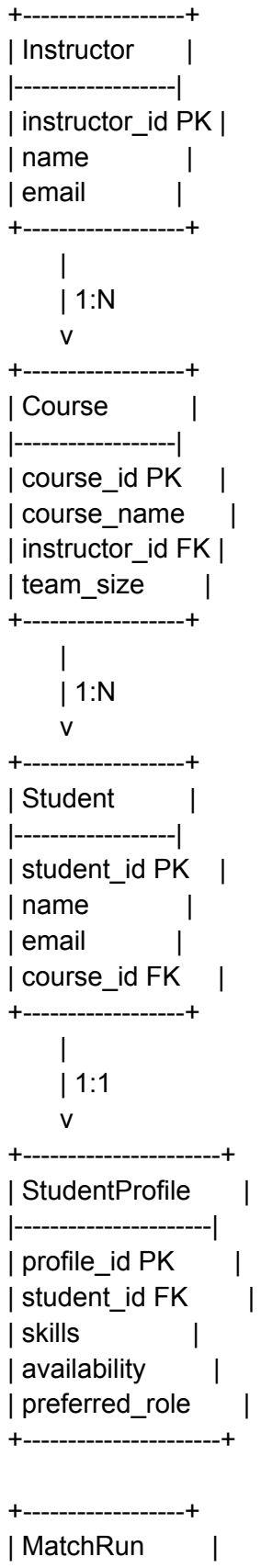
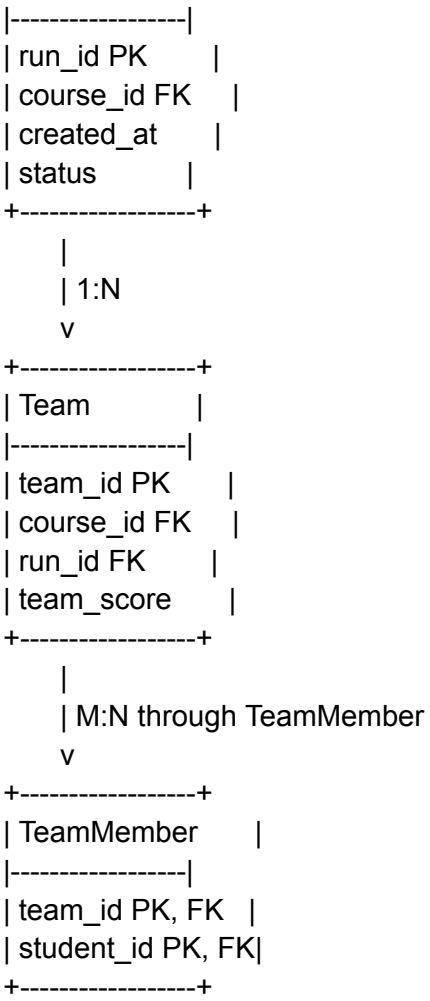
# 7. ER Diagram Table View


Instructor
instructor_id PK
name
email
1:N
Course
course_id PK
course_name
instructor_id FK
team_size
1:N
Student
student_id PK
name
email
course_id FK
1:1
StudentProfile
profile_id PK
student_id FK
skills
availability
preferred_role
MatchRun

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# 8. Short Explanation

The TeamMatch database is designed to support student profile collection, course management, and automated team generation. Students belong to courses and submit profile information such as skills, availability, and preferred role. Instructors manage courses and team-size settings. The system uses this stored data to generate teams during a MatchRun.

The TeamMember table is used to handle the many-to-many relationship between students and teams. This avoids duplicating student information inside the Team table and keeps the database organized. Foreign keys connect students, courses, teams, and match runs, which helps maintain data consistency and supports the main TeamMatch features described in the project requirements.